

Problem Session #02 Simple Database Management

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Task 1: Redesign Data Structure

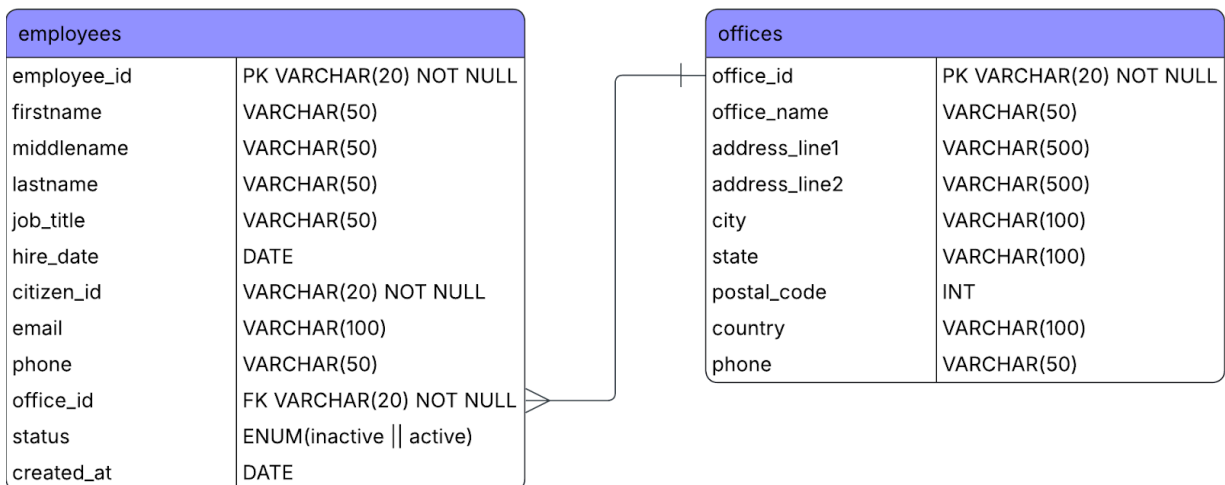
Old Table Design

01: A law firm wants to keep employees and their offices.

1. employees(employee_id, firstname, middlename, lastname, job_title, hire_date, email, phone, office_id, status, created_at)
2. offices(office_id, office_name, address_line1, address_line2, city, state, postal_code, country, phone)

Updated Table Design

1. employees(employee_id, firstname, middlename, lastname, job_title, hire_date, citizen_id, email, phone, office_id, status, create_at)
2. offices(office_id, office_name, address_line1, address_line2, city, state, postal_code, country, phone)



รูปภาพแสดง ER Diagram ข้อ 1

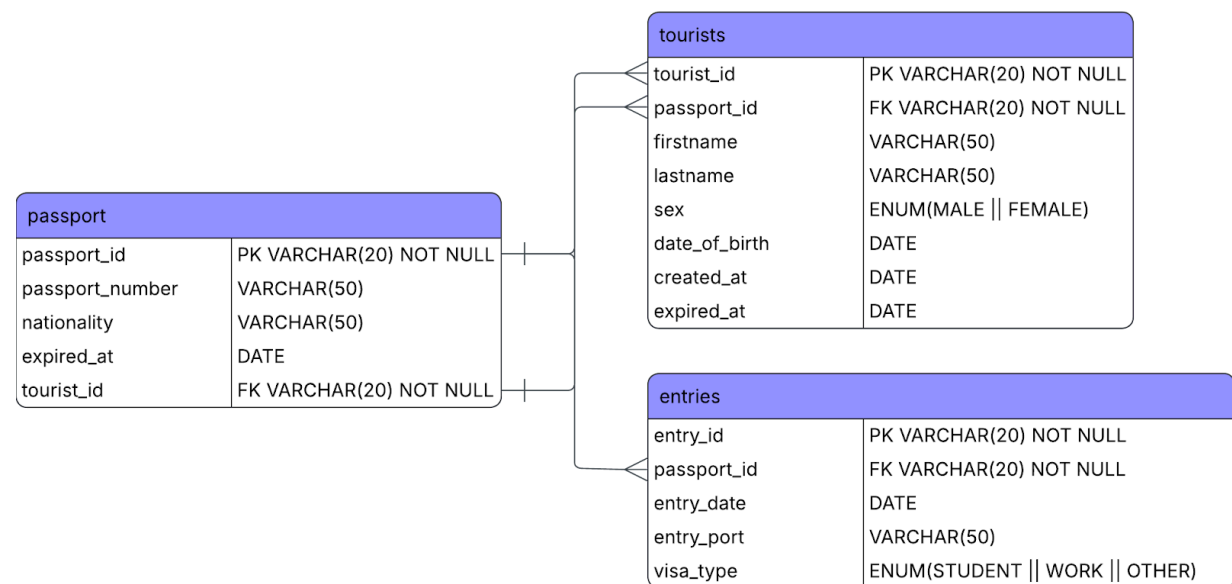
Old Table Design

02: Thai customs want to keep tourists and their passports that they used to enter Thailand.

1. tourists(passport_id, firstname, lastname, sex, date_of_birth, nationality, create_at, expired_at)

Updated Table Design

1. tourists(passport_id, firstname, lastname, sex, date_of_birth, create_at, expired_at)
2. passport(passport_id, passport_number, nationality, expired_at, tourist_id)
3. entries(entry_id, passport_id, entry_date, entry_port, visa_type)



รูปภาพแสดง ER Diagram ข้อ 2

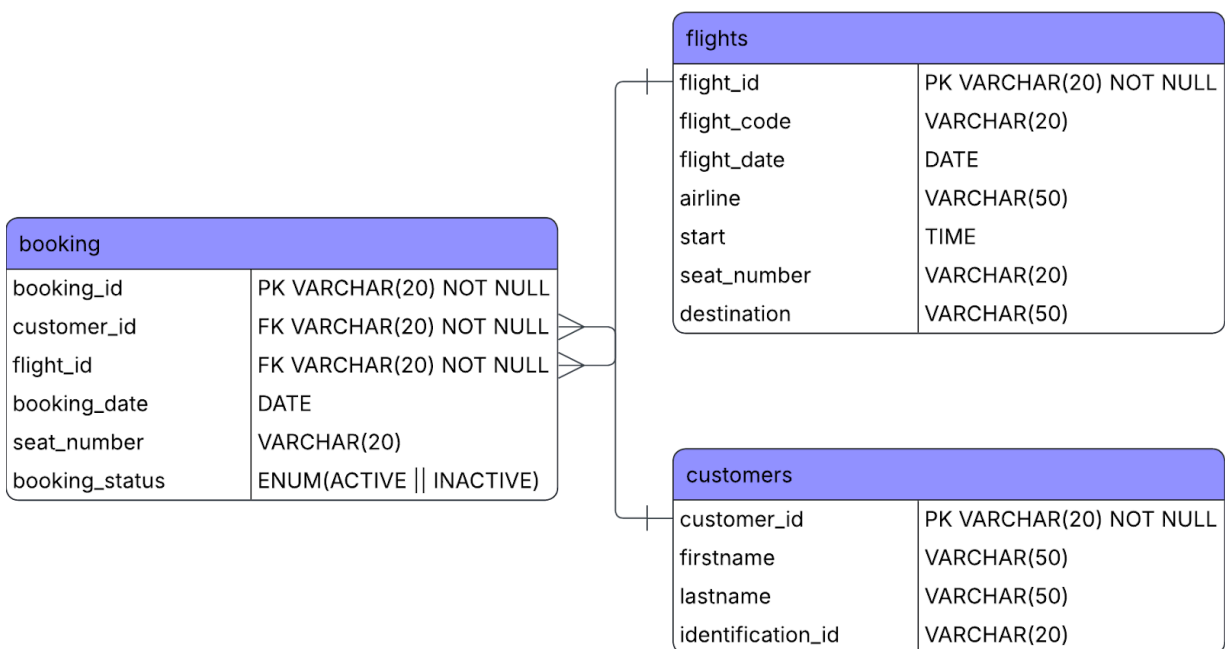
Old Table Design

03: The airline wants to retain its customer base and customer's flight booking.

1. customers(customer_id, firstname, lastname, identification_id)
2. flight(flight_id, flight_code, date, airline, seat, start, destination)

Updated Table Design

1. customers(customer_id, firstname, lastname, identification_id)
2. flight(flight_id, flight_code, flight_date, airline, start, seat_number, destination)
3. booking(booking_id, customer_id, flight_id, booking_date, seat_number, booking_status)



รูปภาพแสดง ER Diagram ข้อ 3

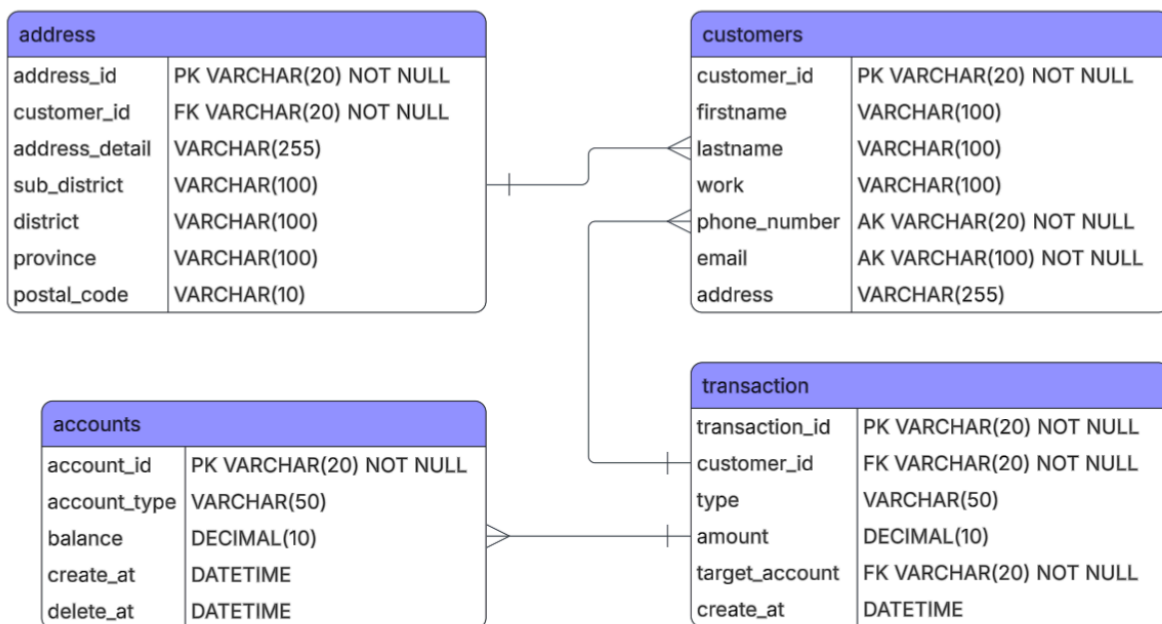
Old Table Design

04: A bank wants to keep customers and their accounts.

1. customers(customer_id, firstname, lastname, phone_number)
2. accounts(account_id, account_type, balance, create_at, delete_at)

Updated Table Design

1. customers(customer_id,firstname,lastname,work,phone_number,email,address)
2. accounts(account_id, account_type, balance, create_at, delete_at)
3. address(address_id, customer_id, address_detail, sub_district, district, province, postal_code)
4. transaction(transaction_id, customer_id, type, amount, target_account, create_at)



รูปภาพแสดง ER Diagram ข้อ 4

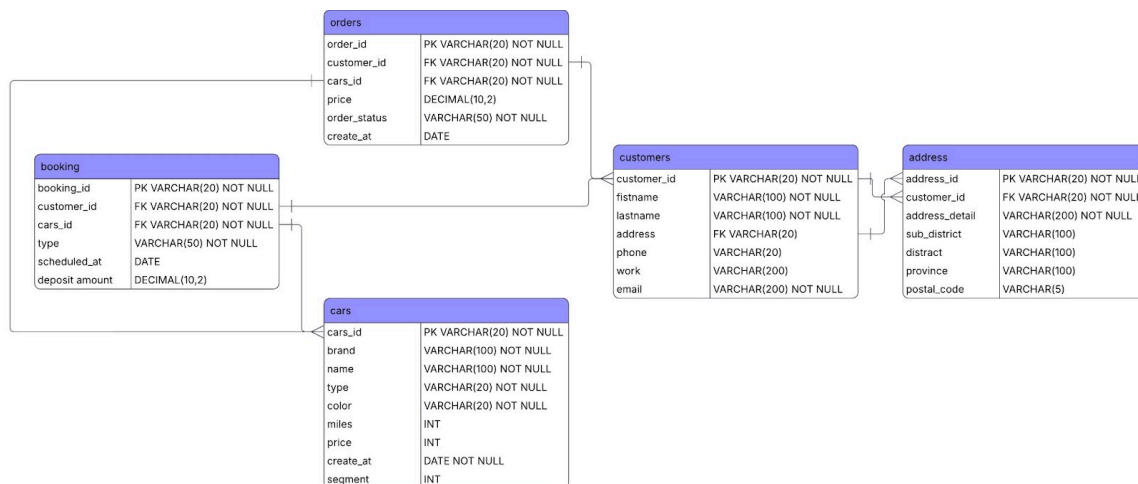
Old Table Design

05: A car dealer that wants to store cars, customers, and orders.

1. cars(cars_id, brand, name, type, miles, price, created_at)
2. customers(customers_id, firstname, lastname, phone, work)
3. orders(order_id, customer_name, cars_name, price, order_status, created_at)

Updated Table Design

1. cars(cars_id, brand, name, type, color, miles, price, created_at, segment)
2. customers(customers_id, firstname, lastname, address, phone, work, email)
3. orders(order_id, customer_name, cars_name, price, order_status, created_at)
4. address(address_id, customer_id, address_detail, sub_district, district, province, postal_code)
5. booking(booking_id, type, scheduled_at, deposit_amount)



รูปภาพแสดง ER Diagram ข้อ 5

Old Table Design

06: A film archive that wants to store movies, actors, and casts.

1. movies(movie_id, title, release_year, genre, duration)
2. actors(actor_id, first_name, last_name, date_of_birth, nationality)
3. casts(cast_id, movie_id, actor_id, role_name)

Updated Table Design

1. movies(movie_id, title, original_title, release_year, release_date, duration_minutes, synopsis, language, country, budget, box_office, rating)
2. actors(actor_id, first_name, last_name, stage_name, date_of_birth, nationality, gender, biography)
3. directors(director_id, first_name, last_name, date_of_birth, nationality)
4. genres(genre_id, genre_name, description)
5. movie_genres(movie_id, genre_id)
6. casts(cast_id, movie_id, actor_id, role_name, billing_order, screen_time_minutes)
7. movie_directors(movie_id, director_id)



รูปภาพแสดง ER Diagram ข้อ 6

Old Table Design

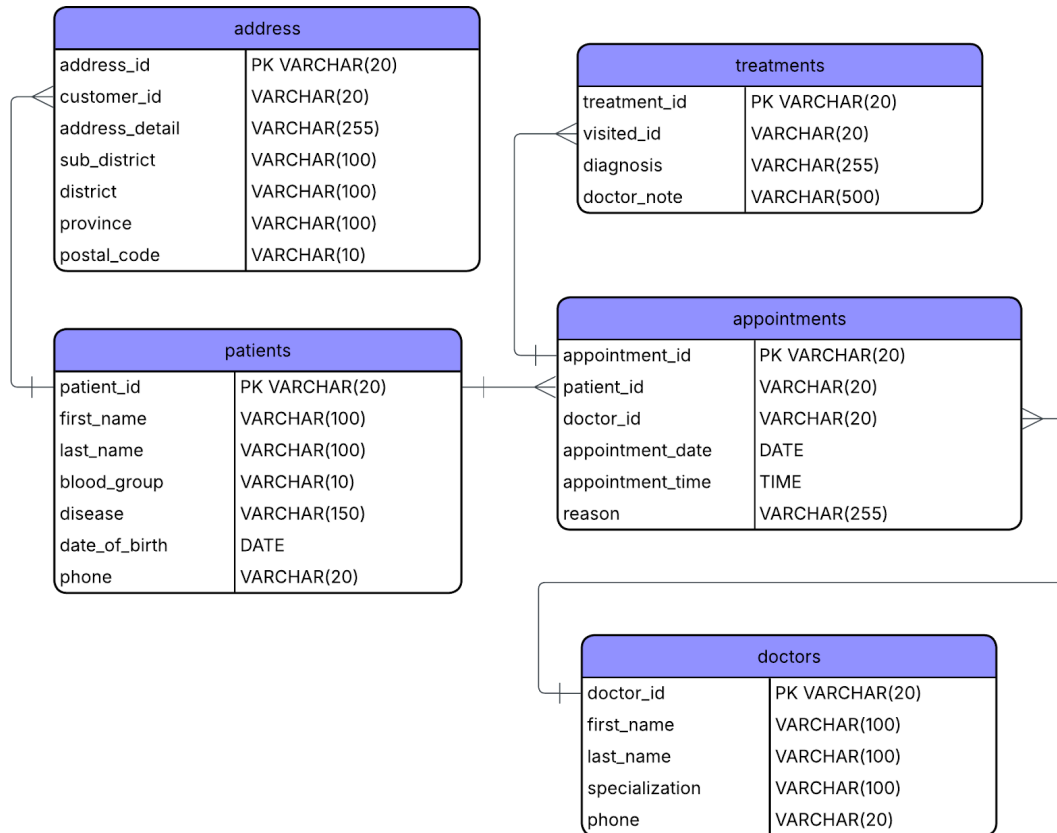
07: A hospital that wants to store patients, doctors, and appointments.

1. patients(patient_id, first_name, last_name, date_of_birth, phone)
2. doctors(doctor_id, first_name, last_name, specialization, phone)
3. appointments(appointment_id, patient_id, doctor_id, appointment_date, appointment_time, reason)

Updated Table Design

1. patients(patient_id, first_name, last_name, blood_group, disease, date_of_birth, phone)
2. doctors(doctor_id, first_name, last_name, specialization, phone)
3. appointments(appointment_id, patient_id, doctor_id, appointment_date, appointment_time, reason)
4. address(address_id, customer_id, address_detail, sub_district, district, province, postal_code)
5. treatments(treatment_id, visited_id, diagnosis, doctor_note)

Updated Table Design



รูปภาพแสดง ER Diagram ข้อ 7

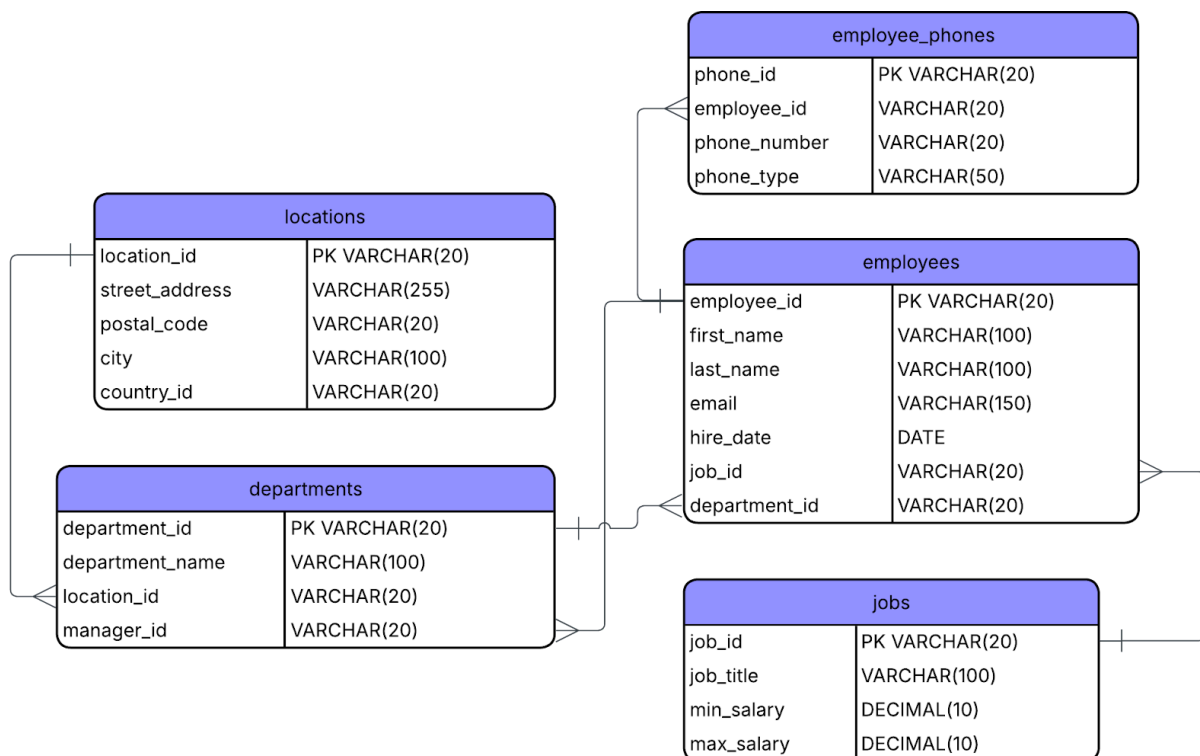
Old Table Design

08: A company that wants to store departments, employees, employee phone numbers

1. departments(department_id, department_name, location)
2. employees(employee_id, first_name, last_name, job_title, department_id)
3. employee_phones(phone_id, employee_id, phone_number, phone_type)

Updated Table Design

1. locations (location_id, street_address, postal_code, city, country_id)
2. departments (department_id, department_name, location_id, manager_id)
3. jobs (job_id, job_title, min_salary, max_salary)
4. employees (employee_id, first_name, last_name, email, hire_date, job_id, department_id)
5. employee_phones (phone_id, employee_id, phone_number, phone_type)



รูปภาพแสดง ER Diagram ข้อ 8

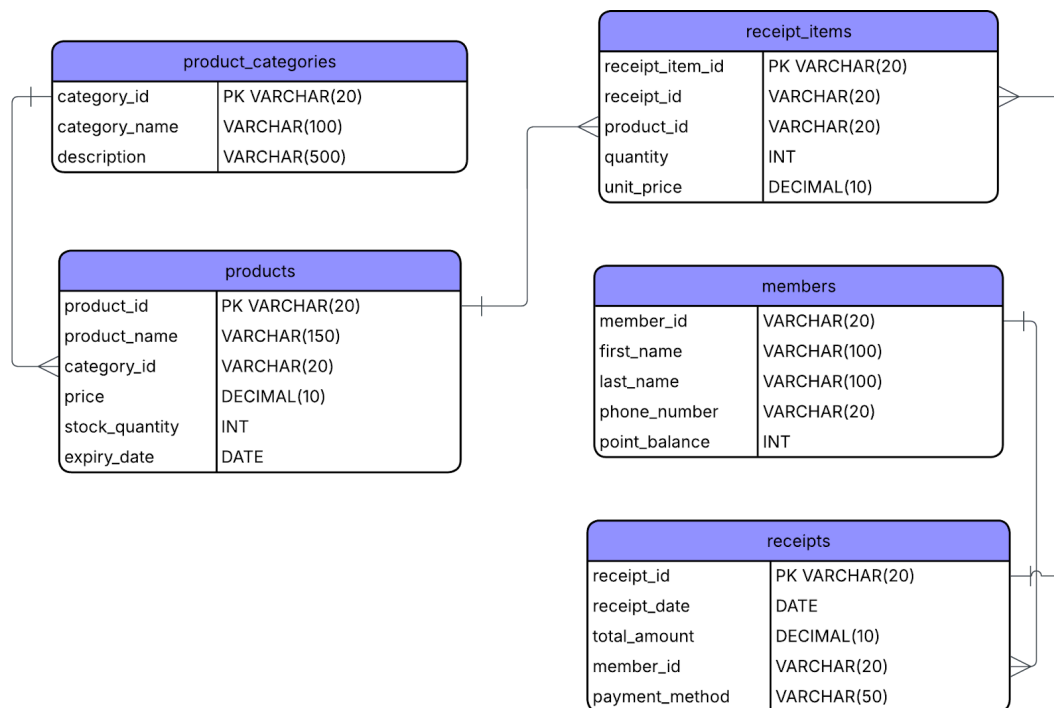
Old Table Design

09: A supermarket that wants to store products, receipts, and reviews.

1. product(product_id,name,ingredient,date of production,expiry date)
2. receipt_item(receipt_item_id, product_id, name_product, price_of_product, date, receipt_id)
3. Receipt(receipt_id, total_price, branch)

Updated Table Design

1. product_categories (category_id, category_name, description)
2. products (product_id, product_name, category_id, price, stock_quantity, expiry_date)
3. members (member_id, first_name, last_name, phone_number, point_balance)
4. receipts (receipt_id, receipt_date, total_amount, member_id, payment_method)
5. receipt_items (receipt_item_id, receipt_id, product_id, quantity, unit_price)



รูปภาพแสดง ER Diagram ข้อ 9

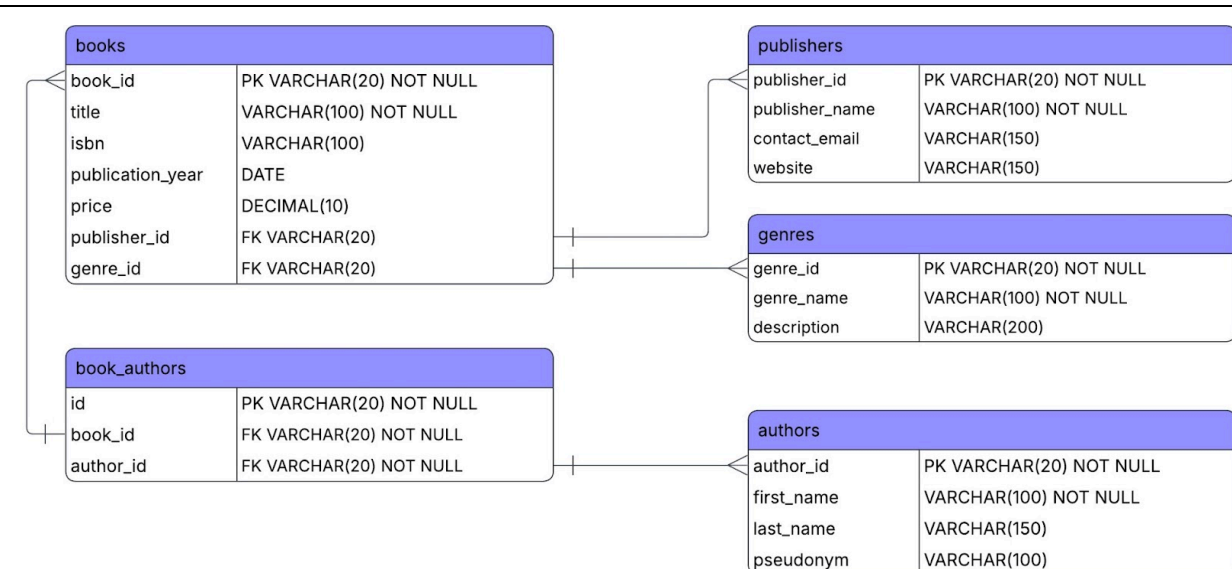
Old Table Design

10: A bookstore wants to keep books and authors of each book.

1. book(book_id, author, type, production_date)
2. author(id, firstname, lastname, phone, address)

Updated Table Design

1. publishers (publisher_id, publisher_name, contact_email, website)
2. genres (genre_id, genre_name, description)
3. books (book_id, title, isbn, publication_year, price, publisher_id, genre_id)
4. authors (author_id, first_name, last_name, pseudonym)
5. book_authors (id,book_id, author_id)



รูปภาพแสดง ER Diagram ข้อ 10

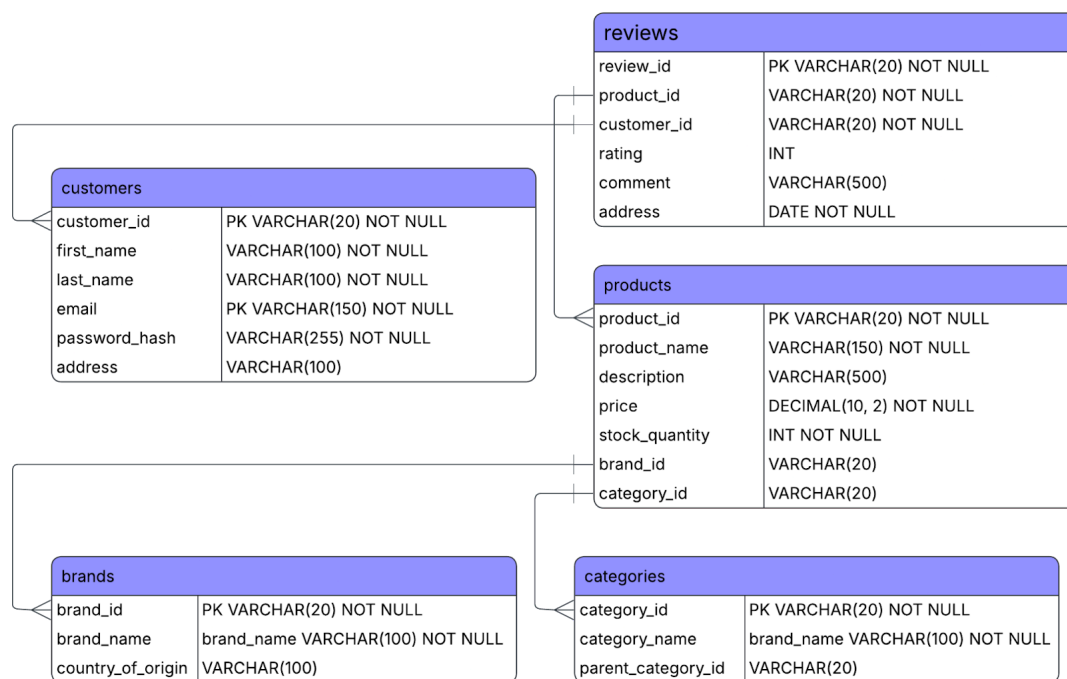
Old Table Design

11: An online store that wants to store customers, products, and reviews

1. customers(customer_id, firstname, lastname, phone, work)
2. products(product_id, name, description, price, stock_quantity)
3. reviews(review_id, customer_id, product_id, rating, comment, review_date)

Updated Table Design

1. brands (brand_id, brand_name, country_of_origin)
2. categories (category_id, category_name, parent_category_id)
3. products (product_id, product_name, description, price, stock_quantity, brand_id, category_id)
4. customers (customer_id, first_name, last_name, email, password_hash, address)
5. reviews (review_id, product_id, customer_id, rating, comment, review_date)



รูปภาพแสดง ER Diagram ข้อ 11

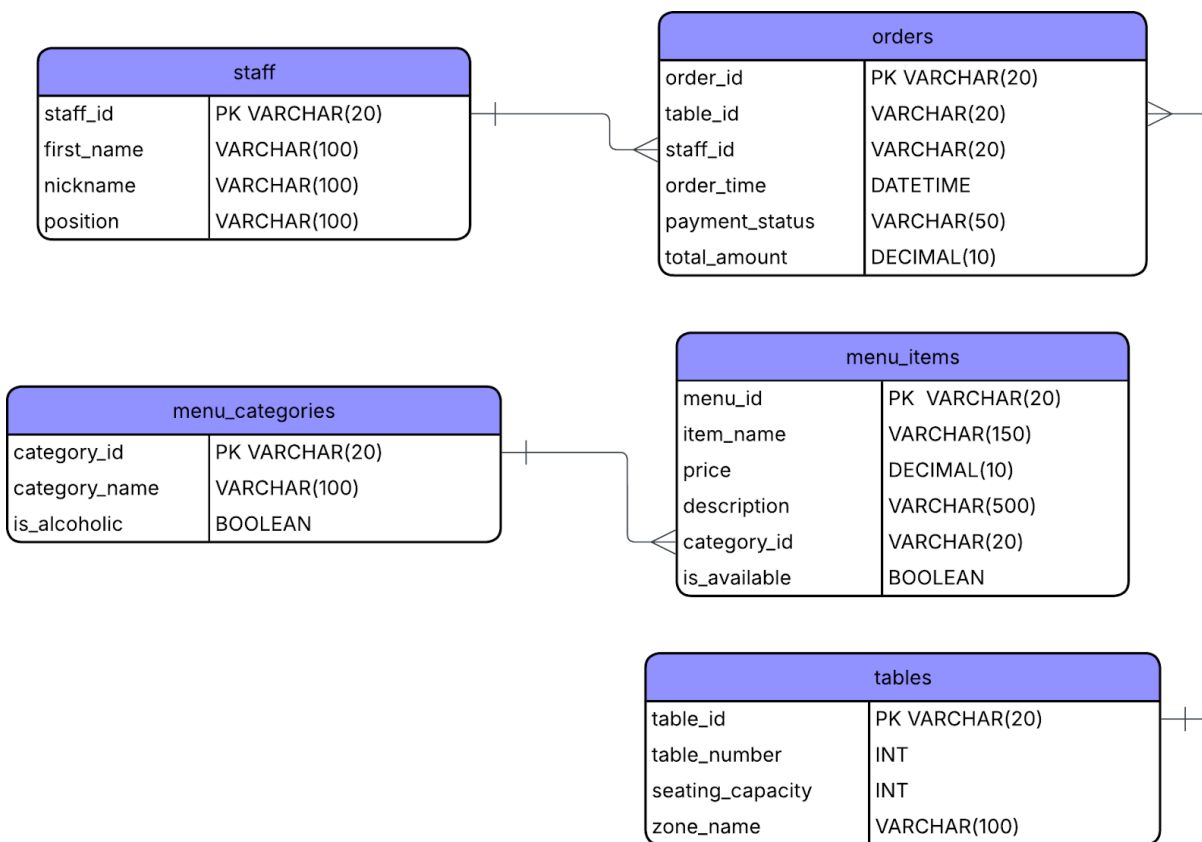
Old Table Design

12: A restaurant wants to keep tables, menus, and order of each table.

1. table(table_id, order, number_people)
2. menu(menu_id, name, price)
3. order(order_id, name_menu, menu_id, table_id)

Updated Table Design

1. menu_categories (category_id, category_name, is_alcoholic)
2. menu_items (menu_id, item_name, price, description, category_id, is_available)
3. tables (table_id, table_number, seating_capacity, zone_name)
4. staff (staff_id, first_name, nickname, position)
5. orders (order_id, table_id, staff_id, order_time, payment_status, total_amount)



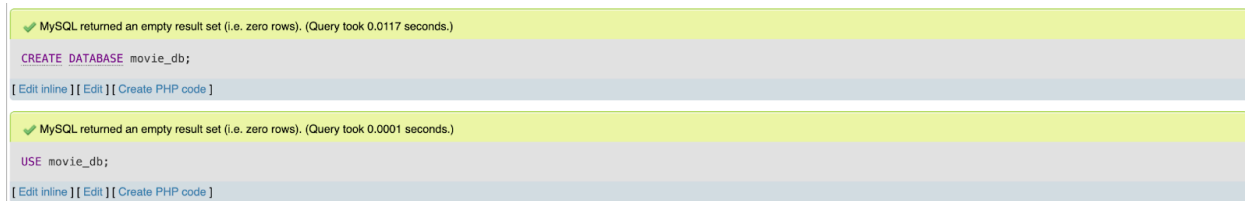
รูปภาพแสดง ER Diagram ข้อ 12

Task 2: Create Database and Insert Data

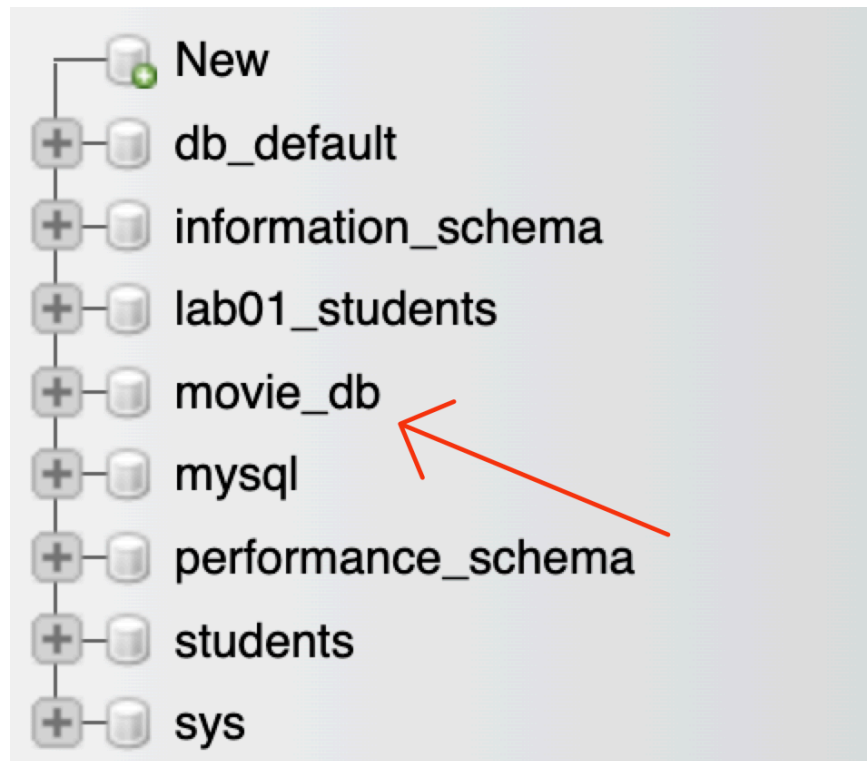
Task นี้ ใช้ข้อมูล Updated Table Design ของข้อ 6 ในการแสดงวิธีทำอย่างละเอียด

Database Creation Evidence

```
CREATE DATABASE movie_db;  
USE movie_db;
```



สร้าง Database “movie_db” ด้วย CREATE DATABASE
และเลือกใช้ Database “movie_db” ด้วย USE



จะเห็นได้ว่าจะมี Database ใหม่เพิ่มเข้ามาชื่อ “movie_db”

Table Creation Evidence

```
CREATE TABLE movies (  
  movie_id VARCHAR(20) NOT NULL,  
  title VARCHAR(100) NOT NULL,  
  original_title VARCHAR(100),  
  release_year INT,  
  duration_minutes INT,  
  synopsis VARCHAR(500),  
  language CHAR(2),  
  country VARCHAR(50),  
  budget BIGINT,  
  box_office BIGINT,  
  rating DECIMAL(3,1),  
  PRIMARY KEY (movie_id)  
);
```

```
CREATE TABLE directors (  
  director_id VARCHAR(20) NOT NULL,  
  first_name VARCHAR(50),  
  middle_name VARCHAR(50),  
  last_name VARCHAR(50),  
  date_of_birth DATE,  
  nationality VARCHAR(50),  
  PRIMARY KEY (director_id)  
);
```

```
CREATE TABLE actors (  
  actor_id VARCHAR(20) NOT NULL,  
  first_name VARCHAR(50),  
  middle_name VARCHAR(50),  
  last_name VARCHAR(50),  
  stage_name VARCHAR(50),  
  date_of_birth DATE,  
  nationality VARCHAR(50),  
  gender ENUM('MALE', 'FEMALE', 'OTHERS'),  
  biography VARCHAR(500),  
  PRIMARY KEY (actor_id)
```

```
);
```

```
CREATE TABLE genres (  
    genre_id VARCHAR(20) NOT NULL,  
    genre_name VARCHAR(50),  
    description VARCHAR(200),  
    PRIMARY KEY (genre_id)  
);
```

```
CREATE TABLE movie_directors (  
    movie_id VARCHAR(20) NOT NULL,  
    director_id VARCHAR(20) NOT NULL,  
    PRIMARY KEY (movie_id, director_id),  
    FOREIGN KEY (movie_id) REFERENCES movies(movie_id),  
    FOREIGN KEY (director_id) REFERENCES directors(director_id)  
);
```

```
CREATE TABLE casts (  
    casts_id VARCHAR(20) NOT NULL,  
    movie_id VARCHAR(20) NOT NULL,  
    actor_id VARCHAR(20) NOT NULL,  
    role_name VARCHAR(100),  
    billing_order INT,  
    screen_time_minutes INT,  
    PRIMARY KEY (casts_id),  
    FOREIGN KEY (movie_id) REFERENCES movies(movie_id),  
    FOREIGN KEY (actor_id) REFERENCES actors(actor_id)  
);
```

```
CREATE TABLE movie_genres (  
    genre_id VARCHAR(20) NOT NULL,  
    movie_id VARCHAR(20) NOT NULL,  
    PRIMARY KEY (genre_id, movie_id),  
    FOREIGN KEY (movie_id) REFERENCES movies(movie_id),  
    FOREIGN KEY (genre_id) REFERENCES genres(genre_id)  
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0444 seconds.)

```
CREATE TABLE movies ( movie_id VARCHAR(20) NOT NULL, title VARCHAR(100) NOT NULL, original_title VARCHAR(100), release_year INT, duration_minutes INT, synopsis VARCHAR(500), language CHAR(2), country VARCHAR(50), budget BIGINT, box_office BIGINT, rating DECIMAL(3,1), PRIMARY KEY (movie_id) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0192 seconds.)

```
CREATE TABLE directors ( director_id VARCHAR(20) NOT NULL, first_name VARCHAR(50), middle_name VARCHAR(50), last_name VARCHAR(50), date_of_birth DATE, nationality VARCHAR(50), PRIMARY KEY (director_id) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0150 seconds.)

```
CREATE TABLE actors ( actor_id VARCHAR(20) NOT NULL, first_name VARCHAR(50), middle_name VARCHAR(50), last_name VARCHAR(50), stage_name VARCHAR(50), date_of_birth DATE, nationality VARCHAR(50), gender ENUM('MALE','FEMALE','OTHERS'), biography VARCHAR(500), PRIMARY KEY (actor_id) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0130 seconds.)

```
CREATE TABLE genres ( genre_id VARCHAR(20) NOT NULL, genre_name VARCHAR(50), description VARCHAR(200), PRIMARY KEY (genre_id) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0170 seconds.)

```
CREATE TABLE movie_directors ( movie_id VARCHAR(20) NOT NULL, director_id VARCHAR(20) NOT NULL, PRIMARY KEY (movie_id, director_id), FOREIGN KEY (movie_id) REFERENCES movies(movie_id), FOREIGN KEY (director_id) REFERENCES directors(director_id) );
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0184 seconds.)

```
CREATE TABLE casts ( casts_id VARCHAR(20) NOT NULL, movie_id VARCHAR(20) NOT NULL, actor_id VARCHAR(20) NOT NULL, role_name VARCHAR(100), billing_order INT, screen_time_minutes INT, PRIMARY KEY (casts_id), FOREIGN KEY (movie_id) REFERENCES movies(movie_id), FOREIGN KEY (actor_id) REFERENCES actors(actor_id) );
```

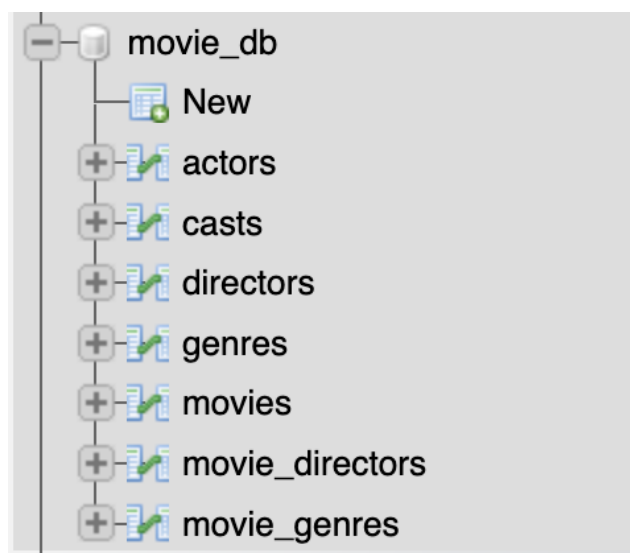
[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0162 seconds.)

```
CREATE TABLE movie_genres ( genre_id VARCHAR(20) NOT NULL, movie_id VARCHAR(20) NOT NULL, PRIMARY KEY (genre_id, movie_id), FOREIGN KEY (movie_id) REFERENCES movies(movie_id), FOREIGN KEY (genre_id) REFERENCES genres(genre_id) );
```

[Edit inline] [Edit] [Create PHP code]

ทำการสร้าง Table movies, directors, actors, genres, movie_directors. casts และ movie_genres ผ่าน CREATE TABLE



จะเห็นได้ว่าจะมี Table ใหม่ถูกสร้างขึ้นภายใต้ Database “movie_db”

Table Description

```
DESCRIBE movies;
DESCRIBE directors;
DESCRIBE movie_directors;
DESCRIBE actors;
DESCRIBE casts;
DESCRIBE genres;
DESCRIBE movie_genres;
```

✔ Your SQL query has been executed successfully. (Query took 0.0005 seconds.)

DESCRIBE movies;

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
movie_id	varchar(20)	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
original_title	varchar(100)	YES		NULL	
release_year	int	YES		NULL	
duration_minutes	int	YES		NULL	
synopsis	varchar(500)	YES		NULL	
language	char(2)	YES		NULL	
country	varchar(50)	YES		NULL	
budget	bigint	YES		NULL	
box_office	bigint	YES		NULL	
rating	decimal(3,1)	YES		NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Movies Table

✔ Your SQL query has been executed successfully. (Query took 0.0003 seconds.)

DESCRIBE directors;

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
director_id	varchar(20)	NO	PRI	NULL	
first_name	varchar(50)	YES		NULL	
middle_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
date_of_birth	date	YES		NULL	
nationality	varchar(50)	YES		NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Directors Table

✔ Your SQL query has been executed successfully. (Query took 0.0002 seconds.)

```
DESCRIBE movie_directors;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
movie_id	varchar(20)	NO	PRI	NULL	
director_id	varchar(20)	NO	PRI	NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Movie_directors Table

✔ Your SQL query has been executed successfully. (Query took 0.0003 seconds.)

```
DESCRIBE actors;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
actor_id	varchar(20)	NO	PRI	NULL	
first_name	varchar(50)	YES		NULL	
middle_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
stage_name	varchar(50)	YES		NULL	
date_of_birth	date	YES		NULL	
nationality	varchar(50)	YES		NULL	
gender	enum('MALE','FEMALE','OTHERS')	YES		NULL	
biography	varchar(500)	YES		NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities Actors Table

✔ Your SQL query has been executed successfully. (Query took 0.0002 seconds.)

```
DESCRIBE casts;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
casts_id	varchar(20)	NO	PRI	NULL	
movie_id	varchar(20)	NO	MUL	NULL	
actor_id	varchar(20)	NO	MUL	NULL	
role_name	varchar(100)	YES		NULL	
billing_order	int	YES		NULL	
screen_time_minutes	int	YES		NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Casts Table

✔ Your SQL query has been executed successfully. (Query took 0.0002 seconds.)

```
DESCRIBE genres;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
genre_id	varchar(20)	NO	PRI	NULL	
genre_name	varchar(50)	YES		NULL	
description	varchar(200)	YES		NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Genres Table

✓ Your SQL query has been executed successfully. (Query took 0.0002 seconds.)

`DESCRIBE movie_genres;`

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

Extra options

Field	Type	Null	Key	Default	Extra
genre_id	varchar(20)	NO	PRI	NULL	
movie_id	varchar(20)	NO	PRI	NULL	

ใช้คำสั่ง Describe เพื่อรีวิว Entities ของ Movie_genres Table

Data Insertion and Selection

-- MOVIES

INSERT INTO movies VALUES

```
('M001', 'Inception', 'Inception', 2010, 148, 'A thief who steals secrets through dream sharing technology.', 'EN', 'United States', 160000000, 829895144, 8.8),
('M002', 'Parasite', 'Gisaengchung', 2019, 132, 'A poor family schemes to infiltrate a wealthy household.', 'KO', 'South Korea', 11400000, 263100000, 8.6),
('M003', 'Interstellar', 'Interstellar', 2014, 169, 'Explorers travel through a wormhole to save humanity.', 'EN', 'United States', 165000000, 677471339, 8.7);
```

-- DIRECTORS

INSERT INTO directors VALUES

```
('D001', 'Christopher', NULL, 'Nolan', '1970-07-30', 'British-American'),
('D002', 'Bong', 'Joon', 'Ho', '1969-09-14', 'South Korean');
```

-- ACTORS

INSERT INTO actors VALUES

```
('A001', 'Leonardo', NULL, 'DiCaprio', NULL, '1974-11-11', 'American', 'MALE', 'Oscar winning actor known for dramatic roles.'),
('A002', 'Joseph', NULL, 'Gordon-Levitt', NULL, '1981-02-17', 'American', 'MALE', 'Actor and filmmaker with diverse roles.'),
('A003', 'Song', 'Kang', 'Ho', NULL, '1967-01-17', 'South Korean', 'MALE', 'Veteran actor acclaimed in Korean cinema.'),
('A004', 'Matthew', NULL, 'McConaughey', NULL, '1969-11-04', 'American', 'MALE', 'Award winning actor known for intense performances.');
```

-- GENRES

INSERT INTO genres VALUES

```
('G001', 'Science Fiction', 'Fictional stories based on advanced science and technology.'),
('G002', 'Thriller', 'Stories focused on suspense and tension.'),
('G003', 'Drama', 'Serious narratives with emotional themes.');
```

```
-- MOVIE_DIRECTORS
```

```
INSERT INTO movie_directors VALUES
```

```
('M001', 'D001'),
```

```
('M003', 'D001'),
```

```
('M002', 'D002');
```

```
-- CASTS
```

```
INSERT INTO casts VALUES
```

```
('C001', 'M001', 'A001', 'Dom Cobb', 1, 120),
```

```
('C002', 'M001', 'A002', 'Arthur', 2, 90),
```

```
('C003', 'M002', 'A003', 'Kim Ki-taek', 1, 110),
```

```
('C004', 'M003', 'A004', 'Cooper', 1, 130);
```

```
-- MOVIE_GENRES
```

```
INSERT INTO movie_genres VALUES
```

```
('G001', 'M001'),
```

```
('G002', 'M001'),
```

```
('G003', 'M002'),
```

```
('G002', 'M002'),
```

```
('G001', 'M003'),
```

```
('G003', 'M003');
```

Show query box

✔ 3 rows inserted. (Query took 0.0191 seconds.)

```
-- MOVIES INSERT INTO movies VALUES ('M001', 'Inception', 'Inception', 2010, 148, 'A thief who steals secrets through dream sharing technology.', 'EN', 'United States', 160000000, 829895144, 8.8), ('M002', 'Parasite', 'Gisaengchung', 2019, 132, 'A poor family schemes to infiltrate a wealthy household.', 'K0', 'South Korea', 11400000, 263100000, 8.6), ('M003', 'Interstellar', 'Interstellar', 2014, 169, 'Explorers travel through a wormhole to save humanity.', 'EN', 'United States', 165000000, 677471339, 8.7);
```

[Edit inline][Edit][Create PHP code]

✔ 2 rows inserted. (Query took 0.0053 seconds.)

```
-- DIRECTORS INSERT INTO directors VALUES ('D001', 'Christopher', NULL, 'Nolan', '1970-07-30', 'British-American'), ('D002', 'Bong', 'Joon', 'Ho', '1969-09-14', 'South Korean');
```

[Edit inline][Edit][Create PHP code]

✔ 4 rows inserted. (Query took 0.0037 seconds.)

```
-- ACTORS INSERT INTO actors VALUES ('A001', 'Leonardo', NULL, 'DiCaprio', NULL, '1974-11-11', 'American', 'MALE', 'Oscar winning actor known for dramatic roles.'), ('A002', 'Joseph', NULL, 'Gordon-Levitt', NULL, '1981-02-17', 'American', 'MALE', 'Actor and filmmaker with diverse roles.'), ('A003', 'Song', 'Kang', 'Ho', NULL, '1967-01-17', 'South Korean', 'MALE', 'Veteran actor acclaimed in Korean cinema.'), ('A004', 'Matthew', NULL, 'McConaughey', NULL, '1969-11-04', 'American', 'MALE', 'Award winning actor known for intense performances.');
```

[Edit inline][Edit][Create PHP code]

✔ 3 rows inserted. (Query took 0.0038 seconds.)

```
-- GENRES INSERT INTO genres VALUES ('G001', 'Science Fiction', 'Fictional stories based on advanced science and technology.'), ('G002', 'Thriller', 'Stories focused on suspense and tension.'), ('G003', 'Drama', 'Serious narratives with emotional themes.');
```

[Edit inline][Edit][Create PHP code]

✔ 3 rows inserted. (Query took 0.0029 seconds.)

```
-- MOVIE_DIRECTORS INSERT INTO movie_directors VALUES ('M001', 'D001'), ('M003', 'D001'), ('M002', 'D002');
```

[Edit inline][Edit][Create PHP code]

✔ 4 rows inserted. (Query took 0.0017 seconds.)

```
-- CASTS INSERT INTO casts VALUES ('C001', 'M001', 'A001', 'Dom Cobb', 1, 120), ('C002', 'M001', 'A002', 'Arthur', 2, 90), ('C003', 'M002', 'A003', 'Kim Ki-taek', 1, 110), ('C004', 'M003', 'A004', 'Cooper', 1, 130);
```

[Edit inline][Edit][Create PHP code]

✔ 6 rows inserted. (Query took 0.0017 seconds.)

■ Console

```
_GENRES INSERT INTO movie_genres VALUES ('G001', 'M001'), ('G002', 'M001'), ('G003', 'M002'), ('G002', 'M002'), ('G001', 'M003'), ('G003', 'M003');
```

[Edit inline][Edit][Create PHP code]

เป็นการ Insert ข้อมูลตัวอย่าง เข้าไปใน Table ที่สร้าง

```

SELECT * FROM movies;
SELECT * FROM directors;
SELECT * FROM actors;
SELECT * FROM genres;
SELECT * FROM movie_directors;
SELECT * FROM casts;
SELECT * FROM movie_genres;

```

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

SELECT * FROM movies;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	movie_id	title	original_title	release_year	duration_minutes	synopsis	language	country	budget	box_office	rating
☐ Edit ☐ Copy ☐ Delete	M001	Inception	Inception	2010	148	A thief who steals secrets through dream sharing L...	EN	United States	16000000	829893144	8.8
☐ Edit ☐ Copy ☐ Delete	M002	Parasite	Chongchung	2019	132	A poor family schemes to infiltrate a wealthy hous...	KO	South Korea	11400000	263100000	8.8
☐ Edit ☐ Copy ☐ Delete	M003	Interstellar	Interstellar	2014	169	Explorers travel through a wormhole to save humani...	EN	United States	165000000	677471339	8.7

Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน Movies Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 1 (2 total, Query took 0.0001 seconds.)

SELECT * FROM directors;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	director_id	first_name	middle_name	last_name	date_of_birth	nationality
☐ Edit ☐ Copy ☐ Delete	D001	Christopher	NULL	Nolan	1970-07-30	British-American
☐ Edit ☐ Copy ☐ Delete	D002	Bong	Joon	Ho	1969-09-14	South Korean

Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน Directors Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 3 (4 total, Query took 0.0001 seconds.)

SELECT * FROM actors;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	actor_id	first_name	middle_name	last_name	stage_name	date_of_birth	nationality	gender	biography
☐ Edit ☐ Copy ☐ Delete	A001	Leonardo	NULL	DiCaprio	NULL	1974-11-11	American	MALE	Oscar winning actor known for dramatic roles.
☐ Edit ☐ Copy ☐ Delete	A002	Joseph	NULL	Gordon-Levitt	NULL	1981-02-17	American	MALE	Actor and filmmaker with diverse roles.
☐ Edit ☐ Copy ☐ Delete	A003	Song	Kang	Ho	NULL	1967-01-17	South Korean	MALE	Veteran actor acclaimed in Korean cinema.
☐ Edit ☐ Copy ☐ Delete	A004	Matthew	NULL	McConaughey	NULL	1969-11-04	American	MALE	Award winning actor known for intense performances...

Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน Actors Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 2 (3 total, Query took 0.0001 seconds.)

SELECT * FROM genres;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	genre_id	genre_name	description
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	G001	Science Fiction	Fictional stories based on advanced science and te...
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	G002	Thriller	Stories focused on suspense and tension.
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	G003	Drama	Serious narratives with emotional themes.

↑ ☐ Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน Genres Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 2 (3 total, Query took 0.0001 seconds.)

SELECT * FROM movie_directors;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	movie_id	director_id
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	M001	D001
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	M003	D001
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	M002	D002

↑ ☐ Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน movie_directors Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 3 (4 total, Query took 0.0001 seconds.)

SELECT * FROM casts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	casts_id	movie_id	actor_id	role_name	billing_order	screen_time_minutes
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	C001	M001	A001	Dom Cobb	1	120
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	C002	M001	A002	Arthur	2	90
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	C003	M002	A003	Kim Ki-taek	1	110
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	C004	M003	A004	Cooper	1	130

↑ ☐ Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน Casts Table ออกมาดู ผ่านคำสั่ง SELECT

Showing rows 0 - 5 (6 total, Query took 0.0001 seconds.)

`SELECT * FROM movie_genres;`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	genre_id	movie_id
☐ Edit Copy Delete	G001	M001
☐ Edit Copy Delete	G002	M001
☐ Edit Copy Delete	G002	M002
☐ Edit Copy Delete	G003	M002
☐ Edit Copy Delete	G001	M003
☐ Edit Copy Delete	G003	M003

☐ Check all
 With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

เรียกข้อมูลใน movie_genres Table ออกมาดู ผ่านคำสั่ง SELECT